

# Polynomial Regression for Geothermal Power Systems

Assignment report | Roll number: BT2024175 | 5 October 2026

**Selected models.** var1: degree-5 LASSO ( $\alpha = 0.01$ ). var2: degree-11 Elastic Net ( $\alpha = 0.001$ , L1 ratio = 0.3). Both choices were made by validation MSE; independent nested cross-validation was used to assess the selection procedure.

## 1. Objective and data

Two independent regression problems were studied: Net Power Score from six turbine operating adjustments (var1), and Thermal Anomaly Score from three spatial coordinates (var2). The aim was to capture nonlinear relationships while controlling overfitting on unseen inputs.

| Property             | var1: turbine        | var2: reservoir       |
|----------------------|----------------------|-----------------------|
| Training / test rows | 1,000 / 1,000        | 1,000 / 1,000         |
| Input variables      | x1 to x6             | x1 to x3              |
| Candidate degrees    | 1 to 10              | 1 to 20               |
| Data checks          | Complete; 0 repeats* | Complete; 12 repeats* |

Complete means no missing values. \*Repeats count rows beyond the first occurrence of an identical training input vector. Training files contain y; test files contain inputs only.

## 2. Polynomial features and preprocessing

For degree  $d$ , all monomials with total degree at most  $d$  were included, including interactions and lower-order terms. For example, a degree-2 expansion of two inputs includes  $x_1$ ,  $x_2$ ,  $x_1^2$ ,  $x_1x_2$  and  $x_2^2$ . A separate constant feature was omitted because the intercept is handled by the regression step.

**Pipeline:** standardize raw inputs → generate polynomial terms → standardize polynomial columns → fit the regularized model.

Each scaler was fitted on the current training fold only. Validation and test rows were transformed using those fitted means and standard deviations. This prevents information from held-out rows influencing training. The supplied inputs already lie within  $[-1, 1]$ ; the first scaling step was retained consistently, while scaling the expanded columns makes regularization less dependent on differences in monomial variance.

With  $p$  inputs, the expansion contains  $C(p + d, d) - 1$  terms. At the maximum degrees this gives 8,007 terms for var1 and 1,770 for var2, both exceeding the 1,000 training observations. This motivates regularization rather than relying on unrestricted high-degree least squares.

### 3. Model selection and validation

Three penalty types were compared. Ridge shrinks all coefficients through an L2 penalty; LASSO can set coefficients exactly to zero through an L1 penalty; Elastic Net combines both. Polynomial columns are often correlated, so the comparison tests whether sparse selection, distributed shrinkage or a mixture is more effective.

| Setting                          | Search and rationale                                                                                                                                                                                 |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Polynomial degree                | Every integer from 1-10 for var1 and 1-20 for var2, following the assignment limits.                                                                                                                 |
| Regularization strength $\alpha$ | Initial grid: $10^k$ , $k = -6, -5, \dots, 6$ . Log spacing covers weak to strong penalties without assuming a useful scale.                                                                         |
| Elastic Net L1 ratio $\rho$      | 0.1, 0.3, 0.5, 0.7, 0.9 and 0.95. LASSO and Ridge separately cover the pure L1 and L2 alternatives.                                                                                                  |
| Local refinement                 | Nine log-spaced $\alpha$ values around the best coarse configuration in each family. Boundary winners extend two decades outward; Elastic Net rechecks all six ratios at its selected coarse degree. |

For  $n$  training rows, the implemented objectives were:

Ridge:  $SSE + \alpha \|\beta\|_2^2$

LASSO:  $SSE/(2n) + \alpha \|\beta\|_1$

Elastic Net:  $SSE/(2n) + \alpha\rho \|\beta\|_1 + \alpha(1 - \rho)\|\beta\|_2^2/2$

Here SSE is the sum of squared training residuals. Because Ridge uses a different loss normalization, numerical  $\alpha$  values are tuned separately and are not directly comparable across families. For LASSO and Elastic Net,  $y$  was centered using the training-fold mean and that mean was added back to predictions; the target was not variance-scaled.

#### Nested cross-validation: five outer folds, three inner folds

**Inner selection.** Within each outer-training partition, the complete degree, model,  $\alpha$  and ratio search used three validation folds. MSE was averaged across folds for each configuration separately, not across model families. The lowest eligible mean MSE selected the configuration.

**Outer evaluation.** The selected pipeline was refitted on the outer-training rows and evaluated once on the held-out outer fold. Repeating this five times separates hyperparameter selection from assessment and reduces the optimism of reporting only the minimum tuning error.

var1 used shuffled KFold. var2 used shuffled GroupKFold with identical coordinate vectors kept together in both loops, so repeated locations could not appear in training and validation simultaneously. Seeds were 175 for inner splits and 176 for outer splits. All final test rows were excluded from fitting and tuning.

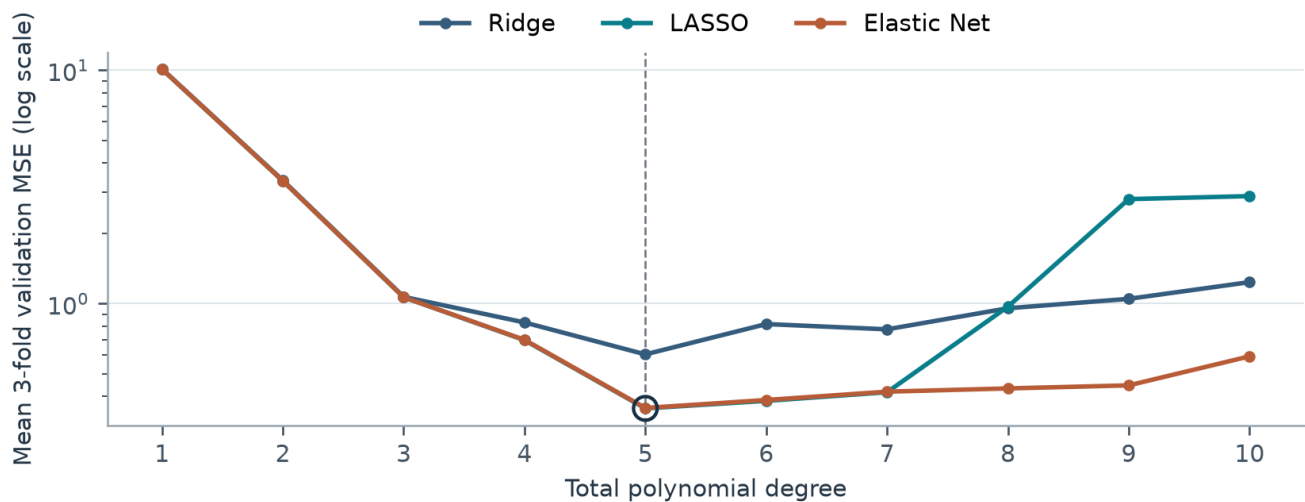
$MSE = SSE/n$  was minimized;  $R^2 = 1 - SSE/SST$  was also reported, with SST measuring variation about the validation-fold mean. Final settings were chosen by a separate three-fold search on all labelled rows, followed by a full-data fit. This search ran before outer evaluation; no outer score changed the final choice. The outer results therefore assess the selection procedure, not a single fixed model trained on all 1,000 rows.

## 4. var1: steam turbine optimization

**Final configuration:** LASSO, degree 5,  $\alpha = 0.01$ .

**Selection CV:** MSE = 0.35519;  $R^2 = 0.96840$ .

**Fitted expansion:** 461 polynomial columns, of which 119 have nonzero coefficients.



**Figure 1.** Lowest fully evaluated, converged three-fold MSE at each degree and model family in the final search, including refinement where performed. The vertical axis is logarithmic; the ring marks the selected model. These are tuning scores, not independent outer-fold scores.

| Best within each family | Degree | $\alpha$ | L1 ratio | CV MSE  |
|-------------------------|--------|----------|----------|---------|
| LASSO                   | 5      | 0.01     | 1.00     | 0.35519 |
| Elastic Net             | 5      | 0.01     | 0.95     | 0.35660 |
| Ridge                   | 5      | 17.78279 | -        | 0.60518 |

### Why this choice?

Validation error fell substantially as the degree increased from 1 to 5, while degrees above 5 did not improve the selected score. Degree 5 therefore provided a better observed balance between capturing nonlinear structure and limiting unnecessary flexibility. LASSO produced the lowest eligible MSE and removed many polynomial coefficients, yielding a sparse representation despite the large expansion.

The near-LASSO Elastic Net configuration was very close: its MSE exceeded LASSO by only about 0.00142. This small difference does not establish a large performance advantage. The predefined minimum-MSE rule determined the final choice, and the substantially higher Ridge MSE suggests that sparse shrinkage was useful for this particular dataset.

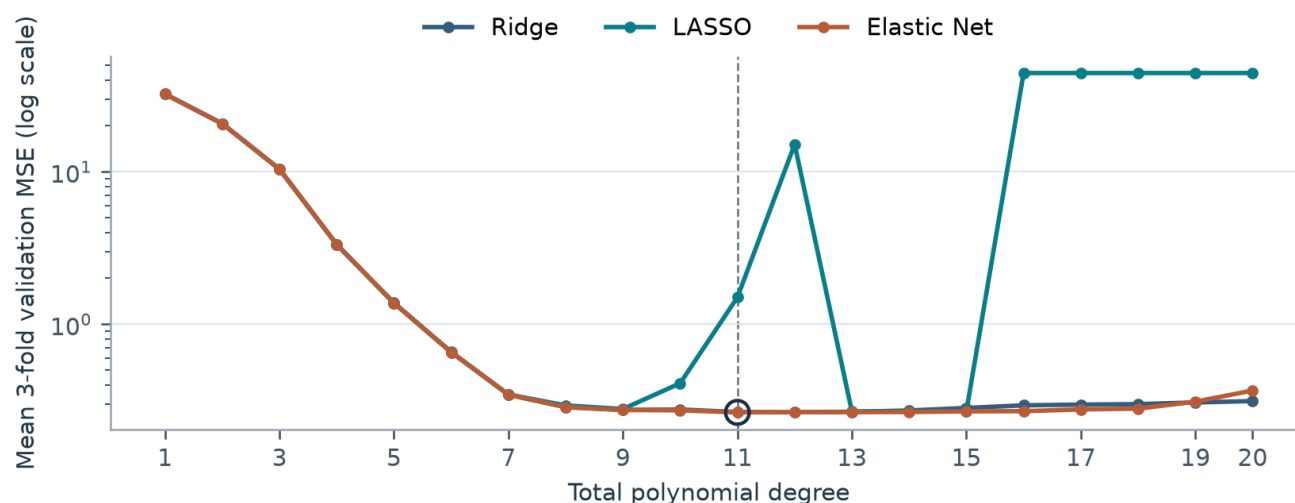
Selection was stable across the independent outer runs: all five chose degree-5 LASSO with  $\alpha = 0.01$ . The mean outer MSE was  $0.33740 \pm 0.04907$  and mean outer  $R^2$  was  $0.96985 \pm 0.00314$ , where  $\pm$  denotes the sample standard deviation across folds. Nonzero coefficients describe this fitted representation; they should not be interpreted as causal effects of plant parameters.

## 5. var2: thermal reservoir mapping

**Final configuration:** Elastic Net, degree 11,  $\alpha = 0.001$ , L1 ratio = 0.3.

**Selection CV:** MSE = 0.26367;  $R^2 = 0.99397$ .

**Fitted expansion:** 363 polynomial columns, of which 233 have nonzero coefficients.



**Figure 2.** Final-search comparison using the definition in Figure 1. Pruned or numerically unfinished candidates are omitted; LASSO spikes should be read alongside the convergence limits in Section 6. The search extends to degree 20; stopping at 10 would have missed the selected configuration.

| Best within each family | Degree | $\alpha$ | L1 ratio | CV MSE  |
|-------------------------|--------|----------|----------|---------|
| Elastic Net             | 11     | 0.001    | 0.30     | 0.26367 |
| Ridge                   | 12     | 1.0      | -        | 0.26427 |
| LASSO                   | 13     | 0.001    | 1.00     | 0.26693 |

### Why this choice?

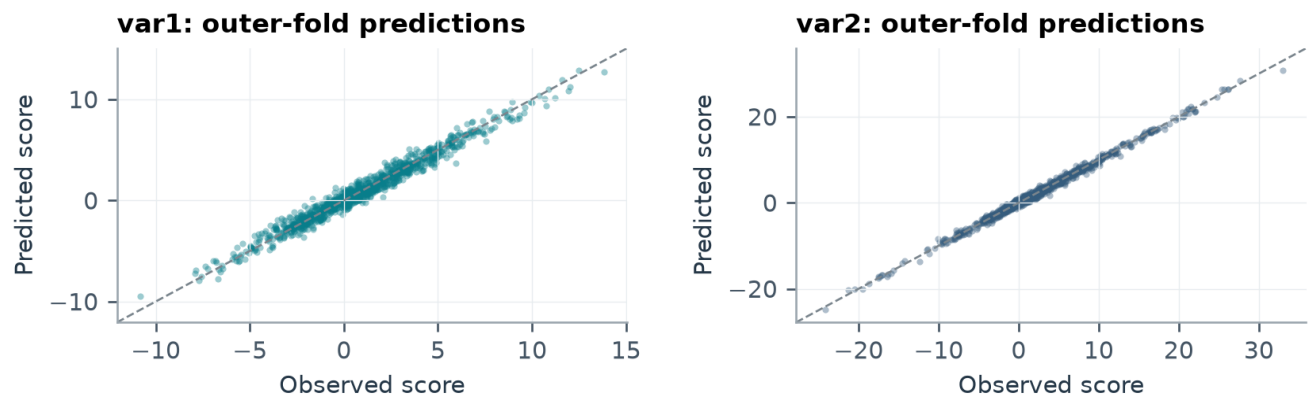
Lower-degree models underfit the observed relationship, while the best scores occurred in a moderate band of higher degrees. Degree-11 Elastic Net achieved the lowest eligible mean MSE. Its L1 component permits coefficient selection and its L2 component distributes shrinkage across correlated terms. The chosen ratio of 0.3 places more weight on the L2 component than on the L1 component in the specified objective.

Ridge at degree 12 was almost tied with the winner: the MSE difference was only about 0.00060. The outer searches selected one LASSO, two Ridge and two Elastic Net models, with degrees ranging from 9 to 13. Thus the data support a useful range of polynomial models rather than a uniquely dominant degree or penalty. Degree 11 is a predictive selection from this search, not an estimate of the true physical polynomial degree.

The mean outer MSE was  $0.27880 \pm 0.03654$  and mean outer  $R^2$  was  $0.99344 \pm 0.00175$ . Grouped validation keeps repeated coordinates out of training when their location is held out. It does not test spatial extrapolation to an entirely new geological region.

## 6. Independent assessment and reproducibility

| Dataset | Mean outer MSE $\pm$ SD | Mean outer $R^2 \pm$ SD |
|---------|-------------------------|-------------------------|
| var1    | 0.33740 $\pm$ 0.04907   | 0.96985 $\pm$ 0.00314   |
| var2    | 0.27880 $\pm$ 0.03654   | 0.99344 $\pm$ 0.00175   |



**Figure 3.** Each point is a training observation predicted when it was held out in an outer fold (1,000 points per dataset). The dashed line denotes perfect agreement. The table reports unweighted fold means; fold SD is descriptive variation, not a confidence interval.

### Numerical checks and limits of the search

Ridge used a shared eigendecomposition across  $\alpha$  values, checked against scikit-learn Ridge. LASSO and Elastic Net used accelerated skglm solvers. Coarse fits were limited to 10 working-set iterations and 500 coordinate epochs per working set; promising unfinished candidates and local refinements received longer budgets. Eligibility required a first-order optimality (KKT) residual at most  $5 \times 10^{-5}$ . Final models used tighter fitting settings and passed this check.

The final-search records contained 36 fully evaluated nonconverged rows for var1 and 311 for var2; these could not win. For var2, another 646 rows were stopped when a partial-error lower bound could not beat a fully evaluated leader. Counts include refinement and retry records, not necessarily unique configurations. Some early var1 checkpoints used a larger solver budget but passed the same eligibility threshold. Consequently, “best” refers to the eligible configurations evaluated with the stated grids, tolerances and refinement rule, not every possible polynomial model.

### Saved predictions and reproducibility

Each chosen pipeline was fitted on all 1,000 labelled rows. Predictions are supplied as **BT2024175\_pred\_var1.csv** and **BT2024175\_pred\_var2.csv**, each with 1,000 rows, one column named y and no index. Original test order and repeated rows were preserved. Reloaded models reproduced both CSVs; all ten outer-fold scores were independently recomputed from saved predictions. No test targets were supplied, so test-set accuracy is unknown.

**GitHub repository:** [github.com/Jos-Samuel/BT2024175-polynomial-regression](https://github.com/Jos-Samuel/BT2024175-polynomial-regression)

Includes train\_polynomial.py, predict.py, pinned requirements, fitted pipelines and search records. Python 3.12, scikit-learn 1.9.1 and skglm 0.5 were used. The README gives training and inference commands; predict.py uses saved models without retraining, and --show-results on the training script prints saved winners.

**Source records:** supplied Polynomial Regression assignment brief; saved final\_search/search\_results.csv, var1/summary.json, var2/summary.json, outer\_predictions.csv and completion\_verification.json. All figures and reported scores were derived from these existing results; no experiments were rerun for this report.